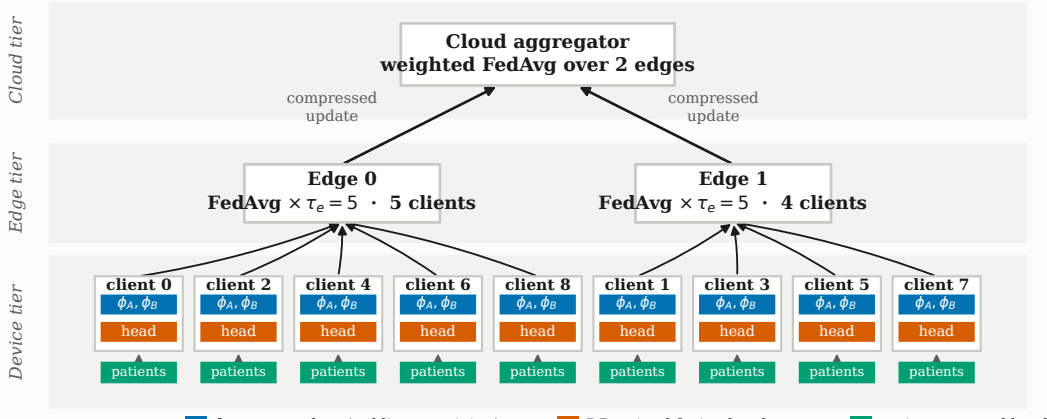
Cloud tier
Edge tier
Device tier
Cloud aggregator
weighted FedAvg over 2 edges
compressed update
compressed update
Edge 0
FedAvg × $\tau_e = 5$ · 5 clients
Edge 1
FedAvg × $\tau_e = 5$ · 4 clients
client 0
client 2
client 4
client 6
client 8
client 1
client 3
client 5
client 7
ϕ_A, ϕ_B
head
patients
frozen encoders (public pretraining)
DP-trained fusion head
patient-grouped local data
data never leaves the client · 6,257 patients, 48,712 windows, partitioned by surgical specialty